INCH-POUND
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SUPERSEDING
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PERFORMANCE SPECIFICATION

HARNESS ASSEMBLY, THERMOCOUPLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four types of thermocouple harness assemblies, hereinafter referred to as the harness.

1.2 Classification. Harnesses are of the following types (see 6.2):

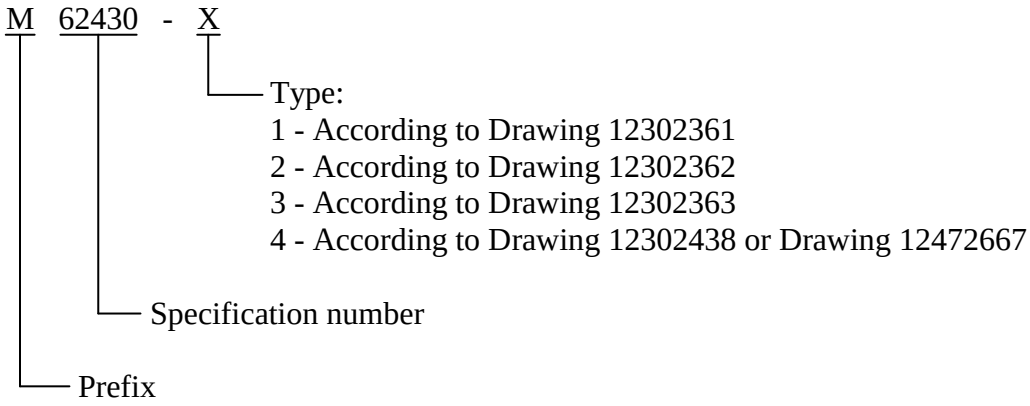
Type I - Differentiating characteristics in accordance with Drawing 12302361 and as specified herein.
Type II - Differentiating characteristics in accordance with Drawing 12302362 and as specified herein.
Type III - Differentiating characteristics in accordance with Drawing 12302363 and as specified herein.
Type IV - Differentiating characteristics in accordance with Drawing 12302438 or Drawing 12472667 and as specified herein.

1.3 Part or identifying number (PIN).

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

COMMERCIAL ITEM DESCRIPTIONS

A-A-59133	-	Cleaning Compound, High Pressure (Steam) Cleaner
A-A-52557	-	Fuel Oil, Diesel

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-PRF-6083	-	Hydraulic Fluid, Petroleum Base, for Preservation and Operation
MIL-PRF-23699	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number 0-156

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	-	Environmental Test Methods
MIL-STD-882	-	System Safety

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DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454	-	General Guidelines for Electronic Equipment
MIL-HDBK-759	-	Human Engineering Design Guidelines

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

U.S. ARMY TACOM DRAWINGS

12284516	-	Rear Engine, Sub-Assembly
12302361	-	Harness Assembly, Thermocouple
12302362	-	Harness Assembly, Thermocouple
12302363	-	Harness Assembly, Thermocouple
12302438	-	Harness Assembly, Electrical, Molded
12472667	-	Harness Assembly, Electrical, Molded

(Copies of these documents are available from DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Zl.4	-	Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted)
ANSI/SAE AS478	-	Identification Marking Methods. (DoD Adopted)

(Copies of these documents are available from ANSI Customer Service Department, 25 W. 43rd Street, 4th Floor, New York, NY 10036 (or website: http://www.ansi.org/) or American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203 (or website: http://www.asq.org/) or Society of Automotive Engineers (SAE) 400 Commonwealth Drive, Warrendale, PA 15096-0001 (or website http://www.sae.org/).)

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ASTM INTERNATIONAL

ASTM E230 - Standard Temperature - Electromotive Force (EMF) Tables for Standardized Thermocouples

(Copies of these documents are available from www.astm.org or ASTM International, P.O. Box C700, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

SAE AMS 3216 - Fluorocarbon (FKM) Rubber High-Temperature - Fluid Resistant Low Compression Set 70 to 80

(Copies of these documents are available from www.sae.org or SAE Customer Service, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Differentiating characteristics. The harnesses shall conform to the following (see 4.5.2):

- Type I - Differentiating characteristics in accordance with Drawing 12302361 and as specified herein
- Type II - Differentiating characteristics in accordance with Drawing 12302362 and as specified herein
- Type III - Differentiating characteristics in accordance with Drawing 12302363 and as specified herein
- Type IV - Differentiating characteristics in accordance with Drawing 12302438 or Drawing 12472667 and as specified herein

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and as noted on the applicable drawing. Materials not designated on the drawing or specifically covered by this specification shall be uniform in quality and free from imperfections or defects which affect their serviceability (see 4.5.1). Thermocouple conductors, sockets, connector inserts, and potting material shall be impervious to the environments and conditions encountered during service.

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3.3.1 Fluid resistance. All materials used in the harness shall be impervious to fluids in accordance with A-A-59133, A-A-52527, MIL-DTL-5624, MIL-PRF-6083, and MIL-PRF-23699 (see 4.5.1).

3.3.2 Corrosion resistance. All external surfaces shall have corrosion resistance at least equivalent to anodized aluminum. Also, there should be no direct contact between dissimilar metals (see 4.5.1).

3.3.3 Thermocouple wire. The thermocouple wire shall be chromel and alumel and shall have an accuracy of ±0.75 percent from 752 to 1832 degrees Fahrenheit (°F) (400 to 1000 degrees Celsius (°C)) (see 4.5.1).

3.3.4 Connector Inserts. Connector insert material shall be fluorocarbon in accordance with SAE AMS 3216 (see 4.5.1). (Applicable to type IV harnesses only.)

3.3.5 Potting. Where applicable, epoxy shall be used in the area where sockets and conductors are connected, to provide additional support (see 4.5.1).

3.3.6 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 4.5.1).

3.4 Design and construction.

3.4.1 Configuration. The harness shall conform to the applicable drawing (see 4.5.1 and 4.5.2).

3.4.1.1 Schematic arrangement. The schematic arrangement for the harness shall be as defined on the applicable drawing (see 4.5.1 and 4.5.2).

3.4.1.2 Interface definition. Electrical and mechanical interfaces shall be as defined on the applicable harness drawing and Drawing 12284516 (see 4.5.1 and 4.5.2).

3.4.2 Weight. The weight of the harness shall not exceed 1.0 pounds (lb) (454 grams (g)) (see 4.5.1 and 4.5.2).

3.4.3 Thermocouple probes. Thermocouple probes shall be closed and ungrounded with a recovery factor of less than 0.75 (see 4.5.1 and 4.5.2).

3.5 Performance.

3.5.1 Individual circuit resistance. The individual circuit resistance between connector contact A and any other contact (except F) at 70 °F (21 °C), shall be in accordance with the following (see 4.5.2):

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a. Type I - 3.19 to 3.67 ohms
b. Type II - 5.04 to 5.80 ohms
c. Type III - 5.37 to 6.17 ohms
d. Type IV - As defined on applicable engineering drawing.

3.5.2 Average circuit resistance. The average circuit resistance between connector contact A and all other contacts (except F) shorted together at 70 °F (21 °C) shall be in accordance with the following (see 4.5.2):

a. Type I - 1.92 to 2.28 ohms
b. Type II - 3.43 to 3.95 ohms
c. Type III - 3.69 to 4.25 ohms
d. Type IV - As defined on applicable engineering drawing.

3.5.3 Insulation resistance. The insulation resistance between the circuits and connector shells and outer braid shall be not less than 25,000 ohms at 250 volts direct current (Vdc) (see 4.5.3).

3.5.4 Output signal. The individual thermocouples shall have an output signal in accordance with ASTM E230 (see 4.5.4).

3.5.4.1 Output signal accuracy. Each individual thermocouple shall produce an output signal within ±11 °F (±6 °C) of the applied temperature, when the thermocouple junction is subjected to an applied temperature of 1112±18 °F (600±10 °C) (see 4.5.4. 1).

3.5.5 Leakage. The full length of sensing probe sheath shall be free of flaws or holes that would allow fuel leakage to enter the inner insulation. The leakage rate shall not exceed 1.83 cubic inches per minute (in^3/min) (3000 cubic centimeters per minute (cm^3/min)) at 200±10 pounds per square inch (psi) (1379±69 kilopascals (kPa)) (see 4.5.2).

3.5.6 Reliability. When operated within the specified environmental and operational stress use envelope, the assembly shall exhibit a Mean-Time-Between-Failure (MTBF) equal to or greater than 10,000 hours (see 4.5.5).

3.6 Environmental conditions.

3.6.1 High temperature. The harness shall meet the specified performance requirements when exposed to the following temperatures (see 4.5.6.1 and 4.5.6.1.1):

a. Electrical connectors - not greater than 400 °F (204 °C)
b. Harness and leads - not greater than 1200 °F (649 °C)
c. Thermocouple probes - 1800±50 °F (982±28 °C)

3.6.2 Low temperature. The harness shall meet the specified performance requirements when exposed to a temperature of -60±10 °F (-51±6 °C) for 4 hours (see 4.5.6.2).

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3.6.3 Altitude. The harness shall meet the specified performance requirements when exposed to an altitude of 1300 feet (ft) (396 meters (m)) below sea level to 10,000 ft (3048 m) above sea level (see 4.5.6.3).

3.6.4 Humidity. The harness shall meet the specified performance requirements, and shall evidence no corrosion or other defects which would affect the function or structural strength, when exposed to 0 to 100 percent (%) relative humidity (see 4.5.6.4).

3.6.5 Salt fog. The harness shall meet the specified performance requirements when exposed to a salt fog (see 4.5.6.5).

3.6.6 Sand and dust. The harness shall meet the specified performance requirements when exposed to blowing dust (fine sand) (see 4.5.6.6).

3.6.7 Fungus. The harness shall not promote the growth of fungus (see 4.5.6.7).

3.6.8 Vibration. The harness shall meet the specified performance requirements when subjected to vibration having levels in accordance with figure 1 from 25 to 500 to 25 Hertz (Hz) (see 4.5.6.8).

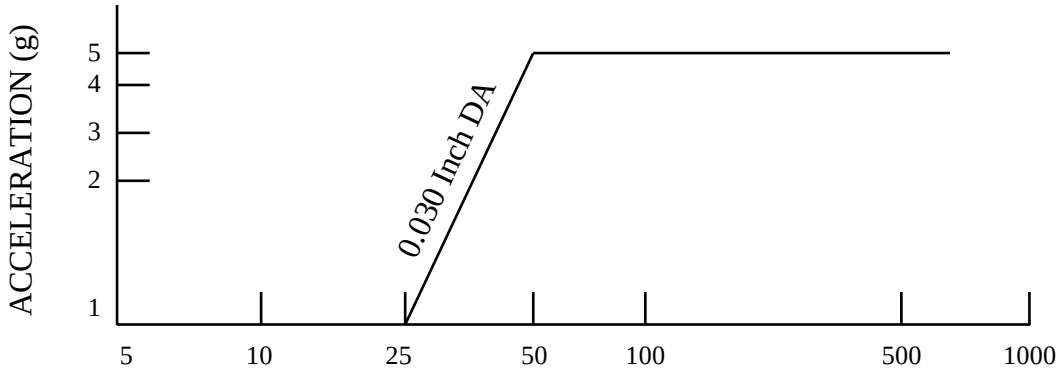


FIGURE 1. Peak vibration envelope.

3.6.9 Gun-firing shock. The harness shall meet the specified performance requirements without malfunction when subjected to shocks of 55±5.5 gravity units (g) (539±54 meters per second squared **(**m/s^2)) for 2.5±0.2 milliseconds (ms) duration in any direction (see 4.5.6.9).

3.6.10 Ballistic shock. The harness shall meet the specified performance requirements without malfunction when subjected to half sine wave shock pulses of 200±20 g (1960±196 m/s^2) for 0.5±0.1 ms duration in any direction (see 4.5.6.10).

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3.6.11 Basic shock. The harness shall meet the specified performance requirements without malfunction when subjected to half sine wave shock pulses of 30±3 g (294±29 m/s^2) for 18 ms duration in any direction (see 4.5.6.11).

3.6.12 Submergence. The harness shall meet performance requirements after submergence in water (see 4.5.6.12).

3.7 Identification marking. Identification marking shall be in accordance with ANSI/SAE AS478 and as noted on the applicable drawing (see 4.5.2).

3.8 Workmanship. MIL-HDBK-454, Guideline 9 (Workmanship) shall be used as guidance for workmanship for the harnesses (see 4.5.7).

3.9 Safety. MIL-STD-882 and MIL-HDBK-759 shall provide design guidance for design safety review, safety hazard levels, safety precedence, analysis, action on identified hazards, and safety testing (see 4.5.8).

4. VERIFICATION

4.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Conformance inspections (see 4.4)
    1. Examination (4.4.2)
    2. Acceptance tests (see 4.4.3)

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature - 73±18 °F (23±10 °C)
b. Pressure - 28.5 +2/-3 inches (in.) (725 +50/-75 millimeters (mm)) mercury (Hg)
c. Relative humidity - 50±30%

4.2.1 Test tolerances. Unless otherwise specified (see 6.2), test conditions, as indicated by instrumentation, shall not deviate from the conditions specified herein by more than the following:

a. Temperature
    Under 100 °F (38 °C) - ±4 °F (±2 °C)
    Over 100 °F (38 °C) - ±4%
b. Pressure - ±5%
c. Relative humidity - ±5%

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d. Vibration amplitude
   Random - ±3 decibels at resonance and ± 1 decibel elsewhere
   Sinusoidal - ±2%
e. Vibration frequency - ±2%
f. Shock - ±5%
g. Acceleration - ±5%
h. Voltage, current, and time
   Value specified as whole number - ±0.5
   Value specified to 1 decimal place - ±0.05
   Value specified to 2 decimal places - ±0.005

4.3 First article inspection. Unless otherwise specified (see 6.2), the Government shall select three harnesses produced under the production contract for first article inspection. First article samples shall be inspected as specified in table I. Approval of the first article samples by the Government shall not relieve the contractor of his obligation to supply harnesses that are fully representative of those inspected as first article samples. Any changes or deviation of the production units from the first article samples shall be subject to the approval of the contracting officer (see 3.2).

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Conformance	
				Examination	Acceptance (100%)
Materials and construction	3.3 thru 3.4.3	4.5.1	X		
Defects (see table II)	3.1, 3.4. 1 thru 3.5.2, 3.5.5, and 3.7	4.5.2	X	X	
Insulation resistance	3.5.3	4.5.3	X		X
Output signal	3.5.4	4.5.4	X		X
Output signal accuracy	3.5.4.1	4.5.4.1	X		X
High temperature	3.6.1	4.5.6.1	X		
Low temperature	3.6.2	4.5.6.2	X		
Altitude	3.6.3	4.5.6.3	X		
Humidity	3.6.4	4.5.6.4	X		
Salt fog	3.6.5	4.5.6.5	X		
Sand and dust	3.6.6	4.5.6.6	X		
Fungus	3.6.7	4.5.6.7	X		
Vibration	3.6.8	4.5.6.8	X		
Gun-firing shock	3.6.9	4.5.6.9	X		
Ballistic shock	3.6.10	4.5.6.10	X		
Basic shock	3.6.11	4.5.6.11	X		
Submergence	3.6.12	4.5.6.12	X		

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4.3.1 Failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection.

4.4 Conformance inspections.

4.4.1 Sampling for examination. Sampling for quality conformance inspection shall be done in accordance with general inspection level II of ANSI/ASQ Zl.4.

4.4.2 Examination. The harnesses shall be examined for the defects listed in table II. One or more defects shall be cause for rejection.

TABLE II. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Assembly incomplete (see 3.4.1).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.4.1).	SIE 1/
103	Weight not as specified (see 3.4.2).	SIE
104	Thermocouple probes not as specified (see 3.4.3).	Visual and SIE
105	Individual circuit resistance not as specified (see 3.5.1).	SIE
106	Average circuit resistance not as specified (see 3.5.2).	SIE
107	Sensing probe sheath leakage greater than specified (see 3.5.5).	SIE
Minor:		
201	Dimensions not affecting interchangeability out of tolerance (see 3.4.1).	SIE
202	Marking improper (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.4.3 Acceptance tests (100%). Each harness shall be subjected to the acceptance tests specified in table I.

4.4.3.1 Failure. Failure of any harness to pass any of the acceptance tests shall be cause for rejection of the failed item.

4.5 Methods of inspection.

4.5.1 Materials and construction. Conformance to 3.3 through 3.4.3 shall be determined by inspection of contractor records providing proof or certification that design, construction,

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processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Defects. Conformance to 3.1, 3.4 through 3.5.2, 3.5.5, and 3.7 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with SIE. Nonconformance shall be cause for rejection.

4.5.3 Insulation resistance. To determine conformance to 3.5.3, the insulation resistance between the individual circuits and connector shells and outer braid shall be measured at an applied potential of 250 Vdc. Nonconformance shall be cause for rejection.

4.5.4 Output signal. To determine conformance to 3.5.4, the output signal of the individual thermocouples shall be measured under the conditions specified in ASTM E230. Nonconformance shall be cause for rejection.

4.5.4.1 Output signal accuracy. To determine conformance to 3.5.4.1, each individual thermocouple junction shall be subjected to an applied temperature of 1112±18 °F (600±10 °C) and the output signal measured. Nonconformance shall be cause for rejection.

4.5.5 Reliability. The satisfactory completion of preproduction testing specified in this document is required for demonstration of the inherent reliability of the product. Usage and test data derived from preproduction and vehicle field use of the same or similar product may be submitted as evidence of demonstrated product reliability, subject to interpretation and approval by the procuring activity. In the event that the test history is insufficient to provide satisfactory evidence of the MTBF specified in 3.5.6, compliance with the product reliability requirement may be determined on the basis of formal reliability analysis and predictions, employing recognized and approved failure rate data sources and prediction procedures as specified by the procuring activity.

4.5.6 Environmental tests. The harness shall be subjected to the environmental tests specified herein. After completion of each environmental test, the performance tests specified in 4.5.3 through 4.5.4.1 and inspections 104 through 106 of table II shall be repeated. Prior to conducting the environmental tests, the harness shall be subjected to an accelerated aging temperature of 392 °F (200 °C) for 168 hours.

4.5.6.1 High temperature. To determine conformance to 3.6.la and 3.6.1b, the harness shall be subjected to the high temperature test of MIL-STD-810, Method 501, Procedure I (Storage). The temperatures specified in 3.6.1 shall apply and the harness shall be maintained at the temperatures for an 8-hour period. Nonconformance shall be cause for rejection.

4.5.6.1.1 Thermocouple probe cycling. To determine conformance to 3.6.1c, upon completion of the high temperature test, the thermocouple probes shall be subjected to not less than 8000 cycles from a room temperature of 77 °F (25 °C) to 1800±50 °F (982±28 °C). Nonconformance shall be cause for rejection.

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4.5.6.2 Low temperature. To determine conformance to 3.6.2, the harness shall be subjected to the low temperature test of MIL-STD-810, Method 502, Procedure I (Storage). The harness shall be placed dry in a temperature chamber with the ambient temperature maintained at -60±10 °F (-51±6 °C) for 4 hours. Nonconformance shall be cause for rejection.

4.5.6.3 Altitude range. To determine conformance to 3.6.3, the harness shall be subjected to an atmospheric pressure of 31.19 in. (792 mm) of Hg (1300 ft (396 m) below sea level), after which the harness shall be subjected to an atmospheric pressure of 20.01 in. (508 mm) Hg (10,000 ft (3048 m**)** above sea level). Nonconformance shall be cause for rejection.

4.5.6.4 Humidity. To determine conformance to 3.6.4, the harness shall be subjected to the humidity test of MIL-STD-810, Method 507, Procedure I (Induced). At the conclusion of the humidity test, the harness shall be dried without forced convection for 12 hours in air at less than 70 °F (21 °C) with not less than 50% relative humidity and then subjected to the performance tests. After this, the harness shall be disassembled and inspected. Nonconformance shall be cause for rejection.

4.5.6.5 Salt fog. To determine conformance to 3.6.5, the harness shall be subjected to the salt fog test of MIL-STD-810, Method 509. Nonconformance shall be cause for rejection.

4.5.6.6 Sand and dust. To determine conformance to 3.6.6, the harness shall be subjected to the sand and dust test of MIL-STD-810, Method 510, Procedure I (Blowing Dust). Nonconformance shall be cause for rejection.

4.5.6.7 Fungus. To determine conformance to 3.6.7, the harness shall be tested in accordance with MIL-STD-810, Method 508. Nonconformance shall be cause for rejection.

4.5.6.8 Vibration. To determine conformance to 3.6.8, the harness shall be mounted using the installed interface configuration for its intended application, or equivalent, and subjected to the test of MIL-STD-810, Method 514, Procedure I (General Vibration), except that the conditions of 3.6.8 shall apply. A 20 minute logarithmic resonant sweep shall be conducted in each of three mutually perpendicular axes, from 25 to 500 to 25 Hz, to identify up to four of the most severe resonant frequencies along each axis. A resonant dwell shall then be conducted for 20 minutes at each of the four resonant frequencies, for a total of 100 minutes in each axis. If less than four resonant frequencies are encountered, the balance of the 100 minutes vibration time in each axis shall be spent cycling the harness over the specified frequency range. Nonconformance shall be cause for rejection.

4.5.6.9 Gun-firing shock. To determine conformance to 3.6.9, the harness shall be mounted using the installed interface configuration for its intended application, or equivalent, and subjected to the test of MIL-STD-810, Method 516, Procedure I (Functional Shock), except that conditions of 3.6.9 shall apply. A total of three half sine wave shock pulses shall be applied

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in both directions along each of three mutually perpendicular axes. Peak amplitude shall be 55±5.5 g (539±54 m/s^2) for 2.5±0.2 ms. Nonconformance shall be cause for rejection.

4.5.6.10 Ballistic shock. To determine conformance to 3.6.10, the harness shall be tested in accordance with MIL-STD-810, Method 516, Procedure I (Functional Shock), except that conditions of 3.6.10 shall apply. The harness shall be mounted normally in a nonoperational mode and subjected to three half sine wave shocks of 200±20 g (1960±196 m/s^2) for 0.5±0.1 ms in each direction along each of three mutually perpendicular axes. Nonconformance shall be cause for rejection.

4.5.6.11 Basic shock. To determine conformance to 3.6.11, the harness shall be mounted using the installed interface configuration for its intended application, or equivalent, and subjected to the test of MIL-STD-810, Method 516, Procedure I (Functional Shock), except that conditions of 3.6.11 shall apply. A total of three half sine wave shock pulses shall be applied in both directions along each of three mutually perpendicular axes. Peak amplitude shall be 30±3g (294±29 m/s^2) for 18 ms. Nonconformance shall be cause for rejection.

4.5.6.12 Submergence. To determine conformance to 3.6.12, all electrical fittings shall be capped and the harness shall be immersed in 70 °F (21 °C) water at a depth of 6 ft (183 centimeters (cm)) for 1 hour. Within 2 hours after completion, the harness shall be subjected to the performance tests of 4.5.3 through 4.5.4.l. Nonconformance shall be cause for rejection.

4.5.7 Workmanship. Workmanship examinations shall be performed at all phases of fabrication, assembly and test, and shall be in accordance with the guidance in MIL-HDBK-454, Guideline 9 (Workmanship) to determine conformance to 3.8.

4.5.8 Safety. A certificate of conformance shall be provided attesting that the assembly safety requirements are in compliance with 3.9.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

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6.1 Intended use. The harness covered by this specification is intended to be used to generate signals required for measurement of gas temperature at the power turbine stator entry on the AGT 1500 gas turbine engine.

6.1.1 Item definition. The harness consists of an integral four probe harness, lead, and output connector assembly. Individual readout signal for each thermocouple is available at the output connector. The individual circuit electrical resistance need not be the same.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, revision, and date of this specification.
b. Title, number, revision, and date of applicable drawings.
c. Type of harness required (see 1.2).
d. If required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2 and 2.3).
e. First article inspection, if required (see 3.2).
f. If inspection conditions are other than as specified (see 4.2).
g. If test tolerances are other than as specified (see 4.2.1).
h. If first article sample size is other than as specified (see 4.3).
i. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Alumel
Chromel
Circuit resistance
Engine, gas turbine

6.4 Supersession data. This specification supersedes Honeywell specification 91547-E2137D, dated 2 September 2002.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing activity:
Army - AT

Review activities:
DLA - GS

(Project 6685-2009-004)

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